

A Cold War Relic or Potential Panacea?: Evaluating the Effectiveness of U.S. Export Control Policy

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Abstract

This paper evaluates a widely held assumption in U.S. national security discourse: that export controls played a key role in the Soviet Union's decline and defeat during the Cold War. Drawing on declassified U.S. intelligence analyses, Department of Defense reports, and congressional hearings, this paper demonstrates that Cold War-era export controls, though extensive in scope, had a very limited effect on preventing the Soviets from accessing Western dual-use and military technologies. Through a sophisticated, government-sponsored effort, the Soviet Union systematically exploited legal loopholes, inconsistent interagency coordination within the U.S. government, lax international enforcement, and other structural deficiencies to bypass the U.S. export regime. In essence, export controls somewhat slowed, but did not prevent Soviet technological progress—and in no way played a significant role in the U.S. victory of the Cold War. This paper offers other explanations for the Soviet Union's technological stagnation. Furthermore, this paper applies these historical insights toward the contemporary U.S.-China great power competition, arguing that export controls should remain a policy tool, but not foundation, of U.S. economic strategy to preserve its technological edge over China.

Introduction

Washington is currently engaged in the single most significant economic war in United States (U.S.) history since the nineteenth century, a comparison that U.S. President Donald Trump himself has invoked (Marshall). Beijing's increased efforts to rival Washington's international influence and military might has increased tensions between the People's Republic of China and the U.S. over the last decade—a tension that has manifested in economic warfare. As the Trump administration has enacted tariffs with a “speed and breadth unmatched in modern history,” Beijing has “responded by imposing controls on

key mineral exports and telegraphing its capacity to throttle supply chains across strategic sectors” (Singh). As the conflict drags longer, policymakers in both Washington and Beijing are increasingly confronted with the reality that “economic warfare is no longer the exception,” but rather, the “main arena of great power competition” (Singh). Though news headlines tend to focus on Washington's inflammatory rhetoric and tariff hikes, a significant aspect of this economic conflict has involved a “high-stakes technological struggle” through the weaponization of export controls (Miller, 302).

U.S. export controls, which have focused on the international transfer of sensitive, dual-use technologies, particularly semiconductors, have been a centerpiece of Washington's national security and foreign policy (Lin). Semiconductors are a type of computing chip embedded in every electronic device and thus are foundational to every tech-based military development in contemporary history. With that said, designing and manufacturing semiconductors is a complicated process, “dominated by a handful of firms in the United States, Japan, and the Netherlands,” all of which have mastered the “highly complex” process of engineering these computer chips (Barbe and Hunt). Notably, though China has developed methods to manufacture some types of semiconductors that power everyday, mundane technologies, Beijing is still reliant on companies based in U.S.-allied nations to source the most complex, high-level semiconductors (Swanson and Mozur). In this way, “U.S. and allied influence over that supply chain” means that export controls can be “a potent weapon against China” (Lind and Mastanduno, 10). Thus, the stakes for such controls to be done in an effective way are immeasurably high. Ultimately, if U.S. export controls can successfully prevent Beijing from developing and acquiring semiconductor chips, China's ability to produce advanced, technology-based military weapons and systems may be severely

limited—a reality with profound economic and military implications.

To this end, political scientist Stephen Walt identifies the question of whether U.S. export controls will slow China's technological advancement as one of the most significant issues facing U.S. policymakers today (Walt). Over the past decade, a small minority of scholars and policymakers have been raising the alarm that the current way of implementing export controls may not be as effective as widely perceived. For example, some argue that export controls should be treated as a "relic" of the Cold War, asserting that in the modern day, such controls are too effective and thus "counterproductive" because they "undercut U.S. tech leadership" (Lewis). Others contend that export controls, as they are currently enforced, are ineffective; however, if the U.S. reevaluates its approach to export controls, Washington may be able to "reestablish" the power of such an economic weapon as an effective instrument in foreign and national security policymaking (Rasser). Still others argue that the U.S. is simply not placing enough pressure on China, and that doubling down on export control enforcement will make them more effective (Amodei and Pottinger). With that said, beyond this minority, a large proportion of policymakers have reached bipartisan consensus that export controls are the new "frontline" for warfighting, a belief predicated on their efficacy (DeBusk). Former National Security Advisor Jake Sullivan, in particular, argues that they are a "strategic asset in the U.S. and allied tool to impose costs on adversaries, and even over time degrade their battlefield capabilities" (The White House Archives, Remarks by National Security Advisor). And policymakers who contend that export controls alone "cannot solve ... all our foreign policy problems or vanquish our adversaries" argue that they can "help ... the U.S. government achieve some important objectives" (Reinsch).

With a few exceptions, much of this modern-day policymaker consensus on the implementation of export controls hinges on the belief that export controls were instrumental to the Soviet Union's collapse, even though export controls were just one of many coercive measures the U.S. took against the Soviet Union during the Cold War. For example, Ramon Marks, a national security lawyer who frequently testifies before Congress on export control issues, writes that "export controls played a critical role in helping the West to prevail over the Soviet Union and its allies in the Cold War" and that "it is not too far of an exaggeration [to argue] that export controls even played the key role, along with the NATO alliance, in ultimately enabling

the Western democracies to prevail over Moscow, its Warsaw Pact allies, and Communist China" (Marks). That said, Marks provides no evidence or evaluative metrics to support his claim about the effectiveness of export controls during the Cold War. Marks is not alone: a broader review of the contemporary literature on this topic reveals that most scholars and policymakers seem to take the efficacy of export controls for granted and fail to identify any definitive measure by which their effectiveness can be evaluated (Wolf; Lichtenbaum).

Given that export controls are central to the U.S. policy toolkit in the current era, it is important to determine whether controls were, in fact, an effective policy measure during the Cold War. Therefore, this paper will evaluate the extent to which export controls were, and remain, an effective means of preserving a technological edge in great power competition. To do so, this paper will examine U.S. intelligence reports to evaluate the efficacy of U.S. export control policy at its most developed point during the Cold War era: during the first Reagan Administration and the 1987 Toshiba-Kongsberg Scandal. In doing so, this paper will demonstrate that U.S. export controls had a negligible impact on the Soviet acquisition of U.S. dual-use and military technologies during the Cold War. To the extent that is possible, this paper will also briefly explore applications of such findings to contemporary U.S. national security policy.

Defining and Exploring the Origins of U.S. Export Control Policy

Export controls are one of several economic warfare mechanisms used by the U.S. More concretely, Kevin Wolf defines export controls as,

The rules governing (1) the export, re-export, and transfer of (2) dual-use, commercial, military, and proliferation-related (3) commodities, technology, software, and services (4) by U.S. and foreign persons (5) to specific destinations, end uses, and end users (6) to accomplish national security or foreign policy—including human rights—objectives. (Wolf)

This paper will focus on export controls for dual-use technologies: those with both civilian and military applications—during the Cold War, U.S. export controls targeted goods beyond such technologies, including oil, gas, and other similar commodities.

In the Cold War context, "anti-communist sentiment" formed Washington's primary ideological basis for export controls, while the "U.S. hegemonic position in the international economy" enabled the

implementation of the export controls (Long, 13). The specific exigence for the export controls was a 1948 American intelligence estimate that warned that the Soviet Union was near to gaining the capacity to “overrun key [U.S. positions] in Europe and Asia” (Cain, 15). The logic behind using export controls to counter growing Soviet conventional military capacity was not grounded in preventing Moscow from gaining nuclear weapons—the estimates evaluated Soviet nuclear weapon potential to be limited at that time—but rather, that “launching a trade war against the East and enlisting the leading European nations in this strategy” would appropriately occupy and divert Soviet attentions from conventional to non-conventional, economic warfare (Cain, 15). Given that initially implementing export controls would require the cooperation of European allies, the Truman Administration tasked the Economic Cooperation Administration (ECA), a U.S. government agency with a presence in Europe because it was originally established to implement aspects of the Marshall Plan, with overseeing the policy’s implementation (Cain, 15).

Export Controls During the Cold War

Unfortunately for Washington, “leading European nations,” including the United Kingdom and France, were less than pleased with the proposed U.S. export controls. Indeed, Western Europeans had a “history of trade with Russia,” a prospect made more tempting by the post-War conditions (Agar, 56). As such, the Europeans saw these export control lists as “unrelated to European recovery and saw them more as a U.S. artifice for drawing them into virtual economic warfare against the Soviets” (Cain, 21). In an attempt to distance themselves from the negotiations, Washington sought to use London as a liaison between the U.S. and European nations. But paradoxically, British–Soviet trade relations subsequently deepened for a short period, as England increased its imports of Russian food and pine timber while exporting British “machinery, tools, metals and engines” because Britain needed Soviet timber for housing, and the export commodities protected the Pound Sterling (Cain, 18; Agar, 56).

Eventually, the U.S. abandoned its reliance on London, and in 1949, instead established a “consultative group of Organisation for European Economic Cooperation countries” to draw up formal embargo lists (Cain, 25). The official steering committee of this consultative group was named the Coordinating Committee for Multilateral Export

Controls (CoCom) (Cain, 25). Codified in the Export Control Act of 1949, CoCom was a secret multilateral institution that included the U.S., Japan, and every NATO member except Iceland, with the express goal of “govern[ing] strategic exports to the Soviet Bloc,” headquartered at an undisclosed location in France (Long, 15; Long, 17). The new export control system was strict, with a case-by-case exemption system, which companies soon sought to exploit. Typically, a “Western European firm would seek an exemption from CoCom restriction, making an argument that downplayed or denied the significance of the goods; the U.S. in turn would monitor and attempt to deny the exemption,” a process that yielded increased disagreement and dissatisfaction amongst the Europeans (Agar, 56). Thus, the establishment of CoCom did not reduce tensions as Washington had hoped.

To appease the Europeans, the U.S. decided it would be best if CoCom’s first legitimate economic target were not Russia, but rather the Chinese and North Korean communist forces with which the U.S. was engaged in conflict on the Korean Peninsula. Under U.S. leadership, CoCom’s initial goal was to cease “any commerce with [the PRC] in order to weaken its economy and its ability to mobilize its strategic resources” by banning exports of items like railroad equipment and rubber shipments to China (Cain, 27; Cain, 28; Cain, 47). At this point, the Department of State relieved the ECA of its role in establishing and implementing export controls, and, as a result, export controls became a more institutionalized, formal tactic of foreign policy (Cain, 28).

But even as Washington increasingly viewed Moscow as its geopolitical rival, U.S. allies in Europe did not. American worries only heightened as 1950 intelligence estimates approximated that conventional war between the U.S. and the Soviet Union was a mere five years away. Washington became increasingly convinced that “industrial strength [would be] the winning factor in war,” and the U.S. discussed how strategic export control policies might incentivize “push the Soviets to arm themselves now” [in 1950] so that Soviet equipment would be obsolete by the time conventional warfare began (Cain, 30). But even as CoCom became more formalized, in April 1950, the U.S. found it difficult to convince its European allies to support U.S. trade embargoes and export control regimes targeting the Soviet Union (Cain, 30). In other words, the goal of U.S. export controls shifted toward a strategy more explicitly geared toward preventing Soviet technological progress. But as the U.S. cut

exports to Europe by 60 percent, James Lay, then-U.S. Secretary of Commerce, lamented that “the beneficiaries [of the U.S. export control policy] have been Russia and manufacturers in Western Europe and the United Kingdom,” while the “sufferers have been American manufacturers and anti-Communist security” (Cain, 30).

Indeed, Paris, in particular, drew Washington’s ire when it signed a Franco-Soviet trade agreement of June 1953, permitting the export of “5,000 ton cargo vessels, 3,000 tons of lead, twenty-five steam boilers and 200 tons of travelling cranes” to the Soviet Union (Cain, 50). The U.S. worried “any form of heavy-duty ship could be converted to military use,” or “its engine removed and put to some other military application,” or even that the lead from such machines could be “used for nuclear shipping,” but Washington “received no support for its prohibitions” (Cain, 51). Indeed, coordinating U.S. allies on export controls remained a problem for Washington throughout the remainder of the Cold War.

Beyond the pressures of the Cold War, regulating “technology transfer is a perennially sensitive issue” for the U.S., because Washington simultaneously faces the “delicate balancing act to promote U.S. exports without jeopardizing national security with many export transactions” (Chapman, 65; Cain, 65). While in the interest of economic profit, the Department of Commerce aims to promote high-tech exports to support U.S. industry, the Department of Defense (DoD) is “concerned that cutting edge technology with military potential might benefit U.S. adversaries” (Chapman, 65). For example, on one hand, the Nixon Administration, extended restrictions on the export of “computer technology, production facilities, specialized programs and comprehensive programming,” as well as technologies that were used to “design, develop, and produce computers,” or any “peripheral storage devices and storage media, displays, high speed memories, and electronic components” in 1974 (The White House Archives, National Security Decision Memoranda 247, 2). On the other hand, the Nixon Administration simultaneously allowed the export of computers with “processing power of 8 million bits per second to be raised to 32 million bits per second” to Soviet countries, and removed export controls on “electromechanical peripheral equipment” (Cain, 65). In essence, the U.S. faced both foreign and domestic challenges to implementing effective export controls during the Cold War.

By the late 1970s, policymakers in Washington

were increasingly raising concerns about the implementation and efficacy of U.S. export controls. For example, in 1977, the House of Representatives Subcommittee on International Economic Policy and Trade of the Committee on International Relations summoned two witnesses from the DoD to address such concerns (U.S. House of Representatives). A 1979 follow-up report published by the General Accounting Office found that “nation’s control system [was] overly complex and that there [was] no effective policymaking structure to clarify export control policies and reconcile them with the need to promote exports” (U.S. General Accounting Office). A re-evaluation of the U.S. export control regime seemed in order.

At the same time, reports from the Central Intelligence Agency (CIA) admitted that “the Soviet military position vis-a-vis the West ha[d] never been stronger” despite the fact that “the Soviet economy [was] under serious strain” (Central Intelligence Agency, *Dependence of Soviet Military Power*, 3). Indeed, though the Soviet Union’s dependence on Western technology had declined from a “near complete reliance” in 1950 to a lesser extent in the 1980s, the lack of Soviet domestic innovation meant that they were still dependent on Western technology and equipment to “develop and manufacture some of their advanced weapons systems more quickly” (Central Intelligence Agency, *Soviet Acquisition of Western Technology*, 5). In other words, Washington had both identified that its existing export control regime was lacking and that the Soviets were still, to a certain extent, reliant on the West in a weak position. Thus, strengthening the U.S. export control regime became increasingly imperative.

The Reagan Administration’s Export Controls on Soviet Technology

The Reagan Administration, which began in the early 1980s, “renewed expressions of international U.S. assertiveness through ... increased defense spending” and built upon the Nixon Administration’s “efforts to balance Soviet power by balancing with China against USSR” (Chapman, 199). With respect to export controls, Washington became increasingly focused on clarifying key terminology, codifying procedures, and, most importantly, “strengthening controls by closing loopholes and improving enforceability” (Chapman, 200). Comparatively to other presidential administrations from the Cold War era, the Reagan Administration developed U.S. export control most significantly.

In November 1981, during the Reagan

Administration's first year, the CIA reported that "existing [export] controls have denied to the USSR very powerful computers but have not prevented the Soviets from illegally acquiring embargoed semiconductor production machinery" and other military-related technologies (Central Intelligence Agency, Dependence of Soviet Military Power, 9). The stakes, the CIA argued, were immeasurably high: that by acquiring "goods and technology" from the West, the Soviet military was improving its "weapon and military support systems," increasing "the efficiency of military and civilian production," and "providing economic gains" that reduced "the burden of defense" (Central Intelligence Agency, Dependence of Soviet Military Power, 1). The CIA recommended "broadening controls over exports of military-related technology-and increasing its efforts to plug leakages," a strategy that the Agency predicted would "slow improvement in the performance of Soviet weapons by the late 1980s or the early 1990s" (Central Intelligence Agency, Dependence of Soviet Military Power, 2). Still, the CIA cautioned that "expanding CoCom controls would force the Soviets to make greater use of non-CoCom suppliers and illegal channels" (Chapman, 112). The CIA further acknowledged that

There is little chance that Western economic [warfare], even if comprehensive and sustained, could markedly affect Soviet military power for the better part of a decade. [Instead,] the main impact of Western economic [warfare] would be to slow qualitative improvements in Soviet weapon systems. (Central Intelligence Agency, Dependence of Soviet Military Power, 11)

In essence, while the Soviet military was certainly dependent on Western investment in research and development, this 1981 report estimated that overall, export controls, among other economic approaches, would likely have a minimal impact on the Soviet military threat toward the U.S.

A second, declassified CIA report dated April 1982 articulated more significant concerns over a "massive, well-planned, and well-managed national-level program approved at the highest party and governmental levels" of the Soviet Union, aimed at facilitating the violation of Western technology export controls (Central Intelligence Agency, Soviet Acquisition of Western Technology, 1). Indeed, the CIA warned that though the U.S. and its allies "traditionally have relied on the technological superiority of their weapons to preserve a credible counterforce to the quantitative superiority of the Warsaw Pact," that such technical superiority was quickly eroding as

the Soviets continued to debut "more and more sophisticated weaponry—weapons that all too often are manufactured with the direct help of Western technology" (Central Intelligence Agency, Soviet Acquisition of Western Technology, 1). This report outlined both legal and illegal mechanisms favored by the Soviets for acquiring goods. The illegal means involved recruiting American-born intelligence assets, the "diversion of controlled technology from legitimate trade channels to proscribed destinations," "in-place diversion, in which legally acquired technology and equipment—in the computer ... are put to military end uses not authorized in export license applications" and industrial espionage, while the legal means involved consulting "open literature," engaging in "legal trade channels, and participating in "student scientific and technological exchanges and conferences" (Central Intelligence Agency, Soviet Acquisition of Western Technology, 2). In other words, though the Soviets certainly spent some effort illegally obtaining access to export-controlled technologies, they also took advantage of several other legal loopholes.

For example, one case study highlighted in the report describes a Soviet scientist who legally participated in the "complex network of international governmental scientific and technical agreements and exchanges that the USSR maintains with the advanced industrial nations" (Central Intelligence Agency, Soviet Acquisition of Western Technology, 4). While on an exchange program in the U.S., this Soviet scientist managed to get assigned to fuel-air explosives research, and "when he finished his US study programs, he almost certainly returned to the USSR to work on related weapons" (Central Intelligence Agency, Soviet Acquisition of Western Technology, 4). Such stories were apparently rather common, since many Soviet bloc scientists who visited the U.S. through such programs participated in high-level "aerohydrodynamic, cryogenic, optic, laser, computer, magnetic bubble computer memory, nuclear, microelectronic, and structural and electronic material areas" before returning home to work on "military-related" programs (Central Intelligence Agency, Soviet Acquisition of Western Technology, 4). In other words, misalignment among U.S. government programs—branches that prioritized cooperation over competition with the Soviets—contributed to a significant loss of U.S. intellectual property through such programs.

Another legal loophole through which the Soviets gained access to sensitive information was the mandated "time out" of classified documents.

Soviet weapons designers and countermeasures experts took advantage of documents automatically declassified every 10 years by the U.S. government's scientific agencies to obtain valuable scientific research and weapons blueprints (Central Intelligence Agency, Soviet Acquisition of Western Technology, 5). Though doing so meant that they were working with information that was a decade or two old, the Soviets were so far behind the U.S. in terms of innovation that these types of loopholes enabled the Soviets to save millions of dollars and hours in research, development, and modernization as well as incorporate countermeasures to Western programs much earlier on into their own weapons systems (Chapman 73).

Yet another legal avenue the Soviets used to subvert CoCom's export control regime was to charter U.S. research firms. Indeed, such firms could "legally purchase controlled U.S. technology and study it without actually violating U.S. export controls unless they attempt to export the equipment or related technical data from the U.S. without a license" (Central Intelligence Agency, Soviet Acquisition of Western Technology, 5). Based on the Soviets' fluctuating research and development priorities, they also used purchasing agents and dummy corporations to infiltrate and purchase materials in the following fields: "computers, materials, semiconductors, communications, navigation and control, vehicular/transportation, lasers and optics, nuclear physics, and microbiology" (Chapman 72). The impact of such intellectual property and material theft can be seen in projects like the Soviet Kama Truck facility (Chapman, 72). The Kama Truck facility was a 7-year construction project that utilized \$150 million in imports from the U.S. and Western Europe to develop automotive production equipment and technology, all of which was later used by Soviet forces in Afghanistan and Eastern Europe (Chapman, 73). By highlighting the deficiencies of the U.S. export control system, this CIA report demonstrates that the many (entirely legal) loopholes within the U.S. export control regime have entirely undermined the purpose of these regulations. The DoD's 1982 report, articulating the Department's budgetary needs for 1983, similarly highlighted such challenges in the U.S. export control regime. Characterizing the "highly coordinated Soviet effort" to circumvent export controls as a "raid on the [U.S.] technology base," the DoD lamented that "under the guise of purchases for benign, civilian uses, the Soviets have obtained a wide range of equipment and technological knowhow critical to

their military program," and "where [the Soviets] failed to get what they want openly, they have resorted to a well-coordinated illegal acquisition program" (U.S. Department of Defense, 80). Furthermore, the DoD highlighted that in spite of this reality, the U.S. export controls regime was too "loose" and enforcement programs too "lax," due to "loopholes in the international control system" and the fact that "clearcut" violations of the controls were not only forgiven, but "forgotten" (U.S. Department of Defense, 81). The result of such a "laissez-faire" attitude, this report explains, is the continuation and increase in scope of the "Soviet raid on [U.S.] technology" (U.S. Department of Defense, 81). In particular, through such means, the Soviets made their "nuclear weapons more accurate," "dramatically improve[d] their airlift capability, and enhanced their "command and control" abilities (U.S. Department of Defense, 81).

The DoD budget request report also highlights CoCom's "serious weaknesses." For example, CoCom's regulations still allowed for the legal export of "shipbuilding, heavy vehicle, and micro-electronic manufacturing processes" to the Soviet Union, saving Soviet scientists "billions of rubles and years of research time" by "avoid[ing] costly mistakes" that beset the U.S. (U.S. Department of Defense, 81). Such efforts, when combined with the Soviets' "comprehensive intelligence system," contributed to the dissemination of several emerging, "high-military value" technologies, including those related to "computer memory, large-scale integrated circuits, genetic engineering, fracture mechanics, and superplasticity" (U.S. Department of Defense, 81). Ultimately, this report catalyzed Congressional approval for a new DoD office, the Defense Technology Security Administration (DTSA), in May of 1985, which became the DoD lead entity for administering export control laws (Chapman, 113).

But even in September of 1985, the CIA produced another report, intended to provide updates on the aforementioned 1982 report, that pointed to continued Soviet technological advancement in key areas. For instance, the report warned that despite the export controls, the Soviets had successfully narrowed the "Western lead in nearly all key technological areas, particularly microelectronics," and that with regard to "materials, explosive, and sensor technologies applicable to deployed tactical forces such as tanks, artillery, and antitank and surface-to-air missiles, the Soviets' technology level is roughly equal to or slightly better than that of the West" (Central Intelligence Agency, Soviet Acquisition of Militarily, 1). The CIA further reported that the Soviets were "the world's

leaders in a few significant fields, such as chemical warfare and in some areas of laser research for future “star wars” applications (Central Intelligence Agency, Soviet Acquisition of Militarily, 1). Notably, the report detailed how the Soviets saved five years and \$55 million in research and development by obtaining documentation on F-14, F-15, and F-18 jet fighters to develop a new radar-guided air-to-air missile system (Central Intelligence Agency, Soviet Acquisition of Militarily, 8). Additionally, acquisition of Western documents “helped the Soviets cut by two years the time spent on researching a new generation of fuzes for munitions with a large kill radius and for self-aiming aviation cluster munitions” (Central Intelligence Agency, Soviet Acquisition of Militarily, 8). And more generally, Western technology “assisted the Soviets in reducing their weapon acquisition cycle by up to two years for research projects in an advanced status” (Central Intelligence Agency, Soviet Acquisition of Militarily, 8). Overall, these gains allowed for “the Soviets to field more sophisticated, versatile, and effective weapons” than they would have otherwise, despite the presence of military and dual-use technology-focused export controls (Central Intelligence Agency, Soviet Acquisition of Militarily, 10).

Though without export controls the Soviet Union would almost certainly have benefited even more significantly from U.S. and Western military technologies, these government reports from the first Reagan administration demonstrate that export controls had a limited impact on preventing the Soviets from acquiring and benefiting from U.S. research and development.

The 1987 Toshiba-Kongsberg Scandal

In 1987, the DTSA (the agency established in response to the DoD’s 1982 report), discovered that the Japanese Toshiba Machine Company was selling “propeller milling machinery to the Soviet Union through Norway’s Kongsberg Vaapenfabrikk Corporation,” violating CoCom restrictions on dual-use exports to the Soviets (Chapman, 74). The scandal was a significant moment of reckoning for the U.S. and CoCom. Clearly, Western export controls were not functioning as intended. The resulting consequences were costly: the Soviets were not only able to build quieter submarines that were harder to detect but also more easily able to attack U.S. nuclear submarines (Chapman, 74; Chapman, 114). As a direct result, where U.S. sensors once detected Soviet submarines 200 miles away, they could detect them

only 10 to 20 miles before visual confirmation (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 16). To this end, then Illinois Senator Alan Dixon lamented that the U.S. no longer had a “monopoly on high technology” and that the Toshiba-Kongsberg incident was a “powerful illustration to [U.S.] allies of the cost of lax enforcement of export controls” (Chapman, 263).

In response to the scandal, the U.S. Congress launched a series of hearings to better understand the deficiencies of U.S. export control policy. A 1987 House of Representatives hearing on the Enforcement of Multilateral Export Controls highlighted the same deficiencies that the CIA had been reporting for years, and warned that “without significant improvements in the enforcement of export controls by CoCom nations, the West [would] continue to undermine its own security” (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 2). At this hearing, Allan Wendt, the Senior Representative for Strategic Technology Policy and State Department representative to CoCom, explained that in the face of a “determined Soviet effort to acquire militarily significant technologies and goods,” CoCom was left trying, with much difficulty, to eliminate the “various differences in national export control systems, many of which were unfortunately based on outdated legislation or informal administrative systems” (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 4). But the resources to “administer the export licensing system and enforce laws and regulations” were lacking in most countries (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 4). Essentially, the allocation of resources to combat such violations was lacking, their efficacy was severely limited.

Stephen Bryen, the Deputy Undersecretary for Trade Security Policy at the Department of Defense, pointed out another of CoCom’s weaknesses at this hearing. For example, Bryen testified that even if other countries sought to enforce existing controls better, the U.S. remained the “single major country to participate [the committee] in the sense of bringing cases to the attention of CoCom for deliberation” (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 8). In other words, mere compliance with cooperation was not sufficient for CoCom to promulgate an effective export control regime; instead, U.S. allies needed to assume responsibility for identifying potential technologies and items for control.

This hearing also publicized the Norwegian

police report that investigated international enforcement of CoCom rulings, which uncovered an even more egregious reality than the scandal itself: that there was “a ten year pattern of illegal sales of high technology equipment to the Soviet Union by companies based in France, Italy, West Germany, and Britain, for use in improving the Soviet Navy and building nuclear weapons” (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 12). In other words, several of the closest allies of the U.S. were both blatantly and consistently violating export controls agreed upon by CoCom. Wendt later clarified in the hearing that some, not all, of these violations were not technically illegal, but rather, took advantage of CoCom policy loopholes that allowed for the licensing of “sophisticated, dual-use technology that has military capability” to an unknown end-user in the Soviet Union (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 14). Regardless, U.S. officials at the hearing bemoaned the consequences of the significant discrepancy between the pledged goals of U.S. allies and their practices. Indeed, Bryen went so far as to argue that “the majority of Soviet defense programs ... benefited in tangible ways from American and other COCOM technologies, and without that technology, they simply wouldn’t be the kind of superpower they are today” (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 24). In other words, the failure to implement export controls properly was a significant factor in the Cold War’s length.

Much of this hearing consisted of U.S. officials blaming their foreign counterparts, and thus, their accounts are inherently biased to deflect responsibility from their departments. With that said, other analyses of the Toshiba-Kongsberg scandal have also identified the U.S. was somewhat at fault. For example, another aspect of the deficiency in export control enforcement that enabled the violation was the confusing and sometimes contradictory nature of export control policy between the State and Defense Departments, and a failure of each agency to “execute due diligence in their export application reviews” (Chapman 201). In other words, a lack of cooperation with key allies and bureaucratic barriers also enabled the Soviet Union to evade export control guidelines.

In response to the scandal, Congress enacted the 1988 Multilateral Export Control Enhancement Act that sanctioned Toshiba, Kongsberg, and “any firms that engaged in militarily significant diversions,” for three years and also granted the U.S. government authority to financially penalize any offenders, with funding to be

used for “U.S. and allied research, development, and procurement costs to counteract the technological advances the Soviet gained by this acquisition and to costs of restoring U.S. military preparedness” (Chapman, 201; Harrison, 2). Despite the House of Representatives and Senate acknowledging that this was a deeply impractical penalization scheme, the legislation passed, reflecting Washington’s “deep anger” over the scandal (Chapman 201). Further, in 1987, the Reagan Administration directed the Department of Commerce to establish the Bureau of Export Administration (BXA), which was tasked with processing export control license applications and enforcing export controls (Chapman, 74). At this point, the Departments of State, Commerce, and Defense each had its own offices related to export controls—a reality that, ironically, increased the difficulty of coordinating enforcement efforts.

Takeaways from the Reagan Administration Era and the Toshiba-Kongsberg Cases

From Washington’s perspective during the Cold War, export controls served their purpose. By early 1981, the CIA was well aware that the main impact of any economic warfare would be to slow down the pace of any qualitative improvements to Soviet weapons systems—in other words, there was no expectation for export controls to “markedly [a]ffect” Soviet military power (Central Intelligence Agency, Dependence of Soviet Military Power, 11). And at the highest levels, export controls did prevent the Soviets from obtaining access to the most powerful U.S.-invented computers and military technology. With that said, beyond these high-level technologies, the Soviets effectively found numerous legal and illegal pathways to acquire export-controlled items. In doing so, the Soviet Union saved tens of thousands of hours and dollars that would otherwise have been invested in research and development of lower-level, but still critical, military technologies and semiconductor production machinery (especially in the context of great power competition).

With that said, contemporary scholars and policymakers seem to have ascribed a different set of aims to these export controls: that they were intended to cripple the Soviet Union and thus played a key role in the outcome of the Cold War—a mistake that conflates causation and correlation. Just because Washington promulgated a wide-ranging export control regime and the U.S. ultimately emerged as the victor of the Cold War does not imply that the controls necessarily played a significant role in the U.S. victory. In fact, this

paper's examination of the evidence overwhelmingly reveals the inefficacy of the U.S. export control regime during the Cold War. In theory, the controls were wide-ranging and restrictive, but in practice, the Soviets engaged in a sophisticated, successful intellectual theft program. To side-step the export control regime, Moscow took advantage of loopholes in the U.S. regime because of limited enforcement capacity or commitment—the Soviets benefited from a lack of policy coordination within the U.S. and a relaxed enforcement approach by Washington's allies internationally. Between the U.S. government's inability to thwart attempts to circumvent CoCom's mandates, the many avenues through which the Soviets could freely obtain research and development intelligence from the U.S., the mandated de-classification of sensitive U.S. documents after a certain number of years, and generally lax, laissez-faire approach to enforcement, the Soviets found avenues to close the conventional military gap between the two blocs more than should have been possible.

Another, more charitable explanation is that, as the scope of the Western export regime expanded during the Cold War, U.S. capacity for enforcement diminished, "encouraging defection" (Lind and Mastanduno, 9). Regardless, the impact of export controls during the Cold War was far more limited than is often acknowledged in the contemporary context.

Implications

Thus, if U.S. export controls were ineffective and not key to the Soviet Union's collapse, then what was? The answer lies in Moscow's self-destructive economic approach and political philosophy, which ultimately played a more significant role in the Soviet Union's collapse than any U.S. policy. For example, though the Soviet Union's researchers had a "scientific understanding...as advanced as that of America's chemists and physicists," top Soviet bureaucrats were more interested in copying stolen U.S. designs for transistors and other high-level technologies, as opposed to investing in domestic research and development programs (Miller, 41). As a result, Soviet scientists and engineers had "little scope for creativity" and started "several years behind the U.S.... and never caught up" (Miller, 43). Furthermore, the structure of Soviet bureaucracy meant that internal promotions favored "better bureaucrat[s]," not those who could devise new products or identify new markets" (Miller, 44). In addition, young Soviet students did not "pursue electrical engineering degrees" or similar career paths because the political-economic system glorified participation in military production, rather than a focus

on innovative, civilian products (Miller, 44). In other words, if Moscow had been more intentionally focused on investing in its own research programs, given the wealth of talent in the Soviet Union, it is entirely possible that it would have developed programs that rivaled or surpassed Washington's. And if the Soviets had combined their intellectual property theft and export control evasion tactics with a fully developed, advanced research program focused on innovation, the outcome of the Cold War may have been vastly different.

Moreover, the Soviet Union was stunted by its refusal to fully embrace technology. Soviet Marshal Nikolai Ogarkov, the chief of staff of the Soviet General Staff, made a startling admission in 1983 to American journalist Leslie Gelb: that Moscow was "afraid of computers" because they would mean a "loss of political control for the Soviet leadership" in terms of information dissemination (Gelb). Ogarkov further elaborated that because Soviet citizens were unfamiliar with technology, it took "years to train Soviet recruits to use [computers], because [recruits] have never used [electronics] before," and that as a result, the Soviets were at a "major strategic disadvantage" and likely to "fall irretrievably behind" the U.S. (Gelb). In other words, by the 1980s, when U.S. export controls against the Soviet Union were just beginning to be implemented in their most intensive form, the Soviet leadership was already identifying that the U.S. was significantly ahead. Thus, regardless of how effective U.S. export controls were, it seems more likely that they merely accelerated a process of Soviet decline that was already in motion, catalyzed by Moscow's policy decisions.

Why, then, do policymakers continue to insist that export controls were effective, despite the plentiful evidence that indicates the contrary? The answer is unclear. It may simply be a case of confusing correlation with causation, and over time, the seemingly self-evident myth of export controls grew beyond control. But another explanation for the continued public praise of export controls may be economic in nature. When the U.S. cut its exports to the Soviet bloc by 60%, it incentivized the creation of a black market, in which any company willing to risk violating CoCom restrictions could effectively monopolize the high-technology and dual-use goods markets in which they operated. This was, in practice, what occurred. As was revealed during the Toshiba-Kongsberg scandal, for nearly the entirety of the Cold War, companies across Western Europe—including Western Germany—were clandestinely selling "high

technology equipment,” used to improve the Soviet Navy and build nuclear weapons, to the Soviets (U.S. House of Representatives, Enforcement of Multilateral Export Controls, 12). In doing so, these Western European companies profited in ways U.S. companies did not. Certainly, export controls needed to be in place, particularly regarding the most advanced types of technology based solely on the U.S. mainland. But it would have been deeply unpopular politically to publicize the fact that companies in allied countries continued to profit where U.S.-based companies could not. It is even entirely possible that a combination of these two explanations explains the bipartisan consensus that Cold War export controls were effective—perhaps, at the time, this political-economic reality drove politicians to avoid publicizing the controls’ failure, and over time, assumptions have replaced facts in public discussion.

Ultimately, if Washington insists on continuing to wield export controls, Washington must first acknowledge the areas where policy change is both necessary and possible. There are some low-hanging fruit the U.S. can implement to ensure the contemporary export control regime is more effective than that of the Cold War era. For example, the declassification of military technology blueprints and documents should only occur when the U.S. is certain that competitors such as China do not already have access to similar technology or weaponry, so as not to undermine existing export control policy. The failure to do so was critical during the Cold War, granting Moscow access to the F-14, F-15, and F-18 designs decades earlier than the Soviets otherwise would have organically innovated to that point. Furthermore, academic exchange programs should not involve the disclosure of high-level, military-related technologies or development. Washington should also be cautious about allowing foreign, state-backed companies to charter research firms in the U.S., particularly those involved in developing high-level, dual-use technologies. And most importantly, the U.S. government must internally align their policies to ensure alignment between the various export control-related divisions of the State, Defense, and Commerce departments. But beyond these areas, it will be difficult for Washington to overcome the other challenges it faced during the Cold War. For example, preventing recruitment of American-born assets, controlling diversions of protected technologies and goods from their intended destinations, and identifying purchasing agents working on behalf of foreign governments are realities that, while the government can be cognizant

of them, are challenging to defend against.

In addition to these areas, U.S. policy should take into account a distinct difference between the Cold War and the contemporary era. Unlike the Cold War, during which the Soviet and Western blocs were, for the most part, separate in terms of economic systems and political interaction, the contemporary state of globalization has “transformed the liberal [economic] order” into one in which economic interdependence is the norm (Farrell and Newman, 44). The intertwined state of global economic networks “increase interdependence between states that were previously relatively autonomous” (Farrell and Newman, 43). As a result, Washington has leveraged this interdependence to wield coercive economic tools, such as sanctions and export controls, with impunity. In addition, more recently, President Trump has pursued an “America First” security strategy that is particularly antagonistic toward U.S. allies (The White House).

European states, in particular, have traditionally been willing to “accept U.S. extraterritorial pressure because of their ‘shared commitment to democratic values... and economic interests’” (Farrell and Newman, 78). With that said, scholars warn that if the U.S. continues to waver from this commitment, it may “face the slow erosion of its ability to... project power globally” (Farrell and Newman, 79). Therefore, it is critical that Washington keep in mind the necessity of allied cooperation to successfully pursue an agenda that involves export controls. Indeed, there may come a point at which U.S. allies no longer wish to cooperate, and given that the lack of allied cooperation in enforcing export controls during the Cold War undermined the overall efficacy of the U.S. regime, exercising restraint or making foreign policy concessions for allies is likely necessary if Washington desires to continue down the path of wielding export controls to protect the U.S. technological edge in relation to China.

Applications to the Current U.S.-China Relationship

As China rose to power in the 2000s, “popular theories about globalization made it sound almost impossible to impose strict controls” (Miller, 201). And with Washington’s focus on befriending its Beijing counterparts, the U.S. didn’t have the “stomach for [another] fight with allies about export controls” (Miller, 201). As a result, even as U.S. intelligence agencies issued warnings, Chinese companies like Huawei took advantage of opportunities in the early 21st century to embed themselves in global markets and networks (Miller, 274). Huawei, in particular, seized a “leading position” in the telecoms and smartphone

markets and, in the 2010s, even began designing its own semiconductor chips (Miller 274; Miller 275). Around the same time, Beijing began implementing a chip subsidy policy in earnest, colluding to push out American chip firms.

It was only in 2018 that the U.S. began to push back, straining economic relations between the two countries. Indeed, as the U.S. fears China's growing ability to develop complex, dual-use technologies, Washington has relied on export controls, with some success, to protect U.S. technological superiority. For example, U.S. President Trump's trade dispute with China escalated in April 2018, when the Department of Commerce concluded that ZTE, a partially state-owned Chinese telecommunications company, had provided false information to U.S. officials (Miller, 302). In response, Washington moved quickly to impose severe export controls on ZTE, cutting off the company's ability to buy U.S. semiconductors, among other items, and imposing a \$2 billion fine (Miller, 302; Freifeld and Lynch). With surprising efficacy, the export controls targeting ZTE led the company to the brink of collapse (Miller, 302). As a result, Washington became convinced that export controls are a "devastatingly powerful weapon" (Miller, 303). In 2019, Washington levied export controls on Huawei, and almost immediately, without "U.S. semiconductors, software and other essential supplies, Huawei, the largest telecommunications-equipment producer in the world, was left struggling to survive" (Palmer). In recent years, Washington has thus continued to weaponize export controls against similar companies such as SMIC, a Chinese state-backed semiconductor foundry, which develops "China's most advanced homegrown artificial intelligence chips in an effort to compete with U.S.-based Nvidia and supply Chinese tech firms with the much-needed chips" (The Associated Press). In the short term, case studies like ZTE's and Huawei suggest that weaponizing economic interdependence through export controls can be an effective form of widening the gap between U.S. and Chinese access to high-level, dual-use technology.

With that said, it is too soon to evaluate the long-term impact of export controls, and there are other reasons to believe that such controls may not be effective in terms of maintaining Washington's long-term technological edge over Beijing. For example, technology and scientific know-how are more widely distributed via the internet, and thus, it is harder for the U.S. to control access to the information on how to develop critical technologies (Rasser, 2). Furthermore, in the age of the internet, intellectual property theft

is exponentially easier than during the Cold War—an act Beijing has proven exceptionally interested in and skilled at. Thus, even if Washington can, in the short term, implement export controls to prevent China from obtaining the materials or parts required to develop advanced technology, Chinese scientists can benefit from years of U.S. research. And unlike the Soviet Union, China is not hindered by self-imposed research and development obstacles. Indeed, the Chinese have demonstrated that they are capable of overcoming steep odds to accomplish unexpectedly innovative, rapid, and efficient technology development (Palmer; Woo and Huang). DeepSeek AI, in particular, completely circumvented the steps thought necessary to create an effective AI tool (Woo and Huang). In the realm of chip development, SMIC, too, has shown a similar penchant for innovation, developing a unique and advanced chip technology despite U.S. restrictions and regulations (Swanson et al.). China has proven it is willing to invest endlessly in developing advanced technology, financially and intellectually, and thus has the capacity to build on U.S. designs in a way the Soviet scientists never could.

Conclusion

Without export controls, the Soviet Union would have taken even greater advantage of U.S. technological advances during the Cold War. This paper does not advocate eliminating export controls from U.S. policy, since they prevented the Soviets from obtaining access to the most sensitive military technologies. But the U.S. was lucky in the sense that the Soviet research and development approach did not prioritize innovation—if the Soviets had more effectively been able to innovate and build upon the intelligence they gathered through the loopholes in the U.S. export control regime, the Cold War may have had a different outcome. In other words, it is difficult to conclude that the export controls themselves were particularly effective, and it is even more difficult to conclude that they were critical to the U.S. Cold War victory.

Therefore, it is imperative that modern policymakers not view export controls as a panacea. Instead, they should be wielded in a cautious and targeted manner to extract the maximum value from their use. Furthermore, they should only be one aspect of the U.S. economic toolkit, not the primary avenue through which the U.S. seeks to protect its technological edge in great power competition against China—an increasingly urgent task, as the U.S. share of global research and development has significantly decreased from 69 percent in 1960 to a mere 28

percent in 2018 (Rasser, 2).

Ultimately, President Trump's November 2025 National Security Strategy document, released just last week, explicated his Administration's desire to reframe the U.S.-China relationship from one of great-power foes to one of economic partnership, "prioritizing reciprocity and fairness to restore American economic independence" (The White House, 20). It remains to be seen whether this will catalyze a significant shift in U.S. economic and export control policy, or if this rhetoric is mere posturing. Regardless, it is unlikely that export controls, particularly those related to the most sensitive types of technologies and dual-use goods, will disappear in the foreseeable future. As such, if the U.S. wants to maintain its technological edge over China by weaponizing export controls, Washington must prioritize cooperation domestically, among federal agencies and policymakers, as well as internationally, between the U.S. and its allies.

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